

Comparative Analysis of Eavesdropping Strategies in BB84 Quantum Key Distribution

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Abstract—Quantum Key Distribution (QKD) enables information-theoretic secure communication by exploiting fundamental principles of quantum mechanics. The BB84 protocol remains the most widely studied and practically deployed QKD scheme. While BB84 is provably secure under idealized assumptions, practical implementations are vulnerable to a variety of eavesdropping strategies that exploit measurement disturbance, device imperfections, and photon source limitations. In this work, we present a comprehensive experimental study of multiple eavesdropping strategies against BB84, including intercept–resend attacks, probabilistic random-basis attacks, entanglement-based probe attacks, and photon number splitting (PNS) attacks. Our analysis spans three complementary experimental layers: large-scale classical Monte Carlo simulation, quantum circuit-level validation using Qiskit Aer, and execution on real IBM Quantum hardware. Using consistent metrics such as quantum bit error rate (QBER), Eve’s information gain, and detectability thresholds, we demonstrate clear trade-offs between information leakage and detectability, highlight the masking effects of realistic noise, and emphasize the critical distinction between protocol-level and implementation-level security in quantum key distribution systems.

Index Terms—Quantum Key Distribution, BB84, Quantum Cryptography, Eavesdropping Attacks, Qiskit, IBM Quantum

I. INTRODUCTION

The security of modern digital communication relies predominantly on classical cryptographic primitives whose security assumptions are computational in nature. The advent of large-scale quantum computing threatens many widely deployed public-key cryptosystems, motivating the exploration of cryptographic techniques with security guarantees rooted in physical law rather than computational hardness. Quantum Key Distribution (QKD) provides such a paradigm by enabling two parties to establish a shared secret key whose security is guaranteed by the fundamental postulates of quantum mechanics.

Among QKD protocols, the BB84 protocol introduced by Bennett and Brassard remains the canonical scheme and forms the basis of most theoretical analyses and practical deployments. BB84 derives its security from the incompatibility of non-orthogonal quantum measurements and the inevitable disturbance caused by eavesdropping. However, while BB84 is provably secure in the asymptotic and idealized setting, real-world implementations deviate from theoretical assumptions due to noise, imperfect photon sources, detector inefficiencies, and hardware constraints.

This work aims to experimentally analyze how different eavesdropping strategies manifest in both idealized and real-

istic settings. Rather than focusing exclusively on analytical security bounds, we emphasize empirical behavior across simulation and hardware execution. By systematically comparing multiple attack models using consistent metrics, we provide insight into the practical detectability and severity of eavesdropping strategies under realistic conditions.

II. BB84 PROTOCOL OVERVIEW

In the BB84 protocol, the sender Alice encodes a random classical bit sequence into quantum states drawn from two mutually unbiased bases: the computational (Z) basis and the diagonal (X) basis. Specifically, bit values are mapped to quantum states $|0\rangle, |1\rangle$ in the Z basis and $|+\rangle, |-\rangle$ in the X basis. The receiver Bob measures each incoming qubit using a randomly chosen basis. After transmission, Alice and Bob publicly announce their basis choices and discard all rounds in which their bases do not match. The remaining bits form the sifted key.

In the absence of noise and eavesdropping, Bob’s measurements perfectly correlate with Alice’s prepared bits. Any attempt by an eavesdropper Eve to extract information necessarily introduces detectable disturbances due to the no-cloning theorem and the collapse of quantum states under measurement. The security of BB84 is thus quantified by estimating the quantum bit error rate (QBER) of the sifted key.

III. THREAT MODEL AND EAVESDROPPING STRATEGIES

We consider an adversary Eve with full access to the quantum channel between Alice and Bob but no access to the authenticated classical channel. Eve’s goal is to maximize her information about the final key while minimizing the disturbance introduced.

A. Intercept–Resend Attack

In the intercept–resend attack, Eve measures each transmitted qubit in a randomly chosen basis and resends a corresponding quantum state to Bob. If Eve’s measurement basis matches Alice’s preparation basis, she obtains the correct bit value without introducing disturbance. If the bases differ, Eve’s measurement outcome is random and collapses the quantum state, introducing errors when Bob later measures in the correct basis.

This attack represents the simplest and most aggressive form of eavesdropping. In ideal BB84, intercept–resend introduces a theoretical QBER of 25

B. Random-Basis (Probabilistic) Attack

The random-basis attack generalizes intercept–resend by allowing Eve to attack only a fraction p of transmitted qubits. With probability p , Eve performs an intercept–resend operation, while with probability $1 - p$, she allows the qubit to pass untouched. This model captures more realistic stealthy adversaries who trade off information gain against detectability.

As p increases, both Eve’s information and the induced QBER increase approximately linearly. At low values of p , Eve can obtain non-negligible information while keeping QBER below typical abort thresholds. This attack highlights that QBER alone may be insufficient to characterize security, particularly in noisy environments.

C. Entanglement-Based Probe Attack

In entanglement-based probe attacks, Eve couples an ancillary probe qubit to Alice’s transmitted qubit via a weak entangling operation. Eve forwards the original qubit to Bob and delays measurement of her probe until after basis reconciliation. By tuning the strength of the interaction, Eve can extract partial information while introducing significantly less disturbance than intercept–resend.

Unlike intercept–resend, Eve does not obtain a classical bit immediately. Instead, Eve’s information is encoded in quantum correlations between her probe and Alice’s qubit. Such attacks are more difficult to detect and require more sophisticated security analysis based on mutual information rather than simple correctness probabilities.

D. Photon Number Splitting (PNS) Attack

Practical BB84 implementations typically use weak coherent pulses rather than ideal single-photon sources. These pulses follow a Poisson photon number distribution, occasionally emitting multiple photons. In a PNS attack, Eve performs a non-demolition measurement of photon number and selectively splits off one photon from multi-photon pulses, forwarding the remaining photons to Bob. Eve can later measure her stored photon in the correct basis without introducing additional QBER.

PNS attacks exploit implementation-level vulnerabilities and cannot be detected by QBER estimation alone. Our study analyzes PNS attacks using classical simulation, highlighting the necessity of countermeasures such as decoy-state protocols.

IV. EXPERIMENTAL METHODOLOGY

Our experimental framework consists of three complementary layers designed to capture different aspects of BB84 security.

A. Classical Monte Carlo Simulation

Large-scale classical simulations were used to model BB84 and eavesdropping strategies with up to 10^5 transmission rounds. This layer provides statistically stable estimates of QBER, Eve’s information gain, and key length under varying attack parameters and noise levels.

B. Quantum Circuit Simulation

Quantum circuit-level experiments were conducted using Qiskit Aer to explicitly model state preparation, measurement, and entanglement. These simulations validate that the observed trends are consistent with quantum mechanical behavior at the gate level.

C. Real Quantum Hardware

Selected experiments were executed on IBM Quantum hardware using the SamplerV2 primitive. Due to hardware noise and limited shot budgets, these experiments serve as qualitative validation rather than statistical benchmarks.

V. EVALUATION METRICS

The security of a quantum key distribution protocol cannot be assessed using a single scalar quantity. Instead, multiple complementary metrics are required to capture different aspects of protocol correctness, eavesdropper capability, and detectability. In this work, we adopt a set of evaluation metrics that are standard in the QKD literature but are interpreted empirically through extensive simulation and hardware experimentation.

A. Quantum Bit Error Rate (QBER)

The Quantum Bit Error Rate (QBER) is the primary observable used in BB84 to detect the presence of an eavesdropper. It is defined as the fraction of mismatched bits between Alice and Bob after basis reconciliation:

$$\text{QBER} = \frac{1}{|K|} \sum_{i \in K} \mathbf{1}(A_i \neq B_i), \quad (1)$$

where K denotes the set of indices corresponding to the sifted key, A_i and B_i are Alice’s and Bob’s bits respectively, and $\mathbf{1}(\cdot)$ is the indicator function.

In ideal BB84 operation without noise or eavesdropping, the QBER approaches zero asymptotically. However, both eavesdropping and physical noise contribute to non-zero QBER. A key challenge in practical QKD is distinguishing between benign noise-induced errors and adversarially induced disturbances. Throughout this work, QBER serves as the principal metric for eavesdropper detectability.

B. Eavesdropper Information Gain

While QBER quantifies disturbance, it does not directly measure how much information an adversary has obtained. We therefore also evaluate Eve’s information gain. For attacks in which Eve performs explicit measurements and stores classical outcomes (intercept–resend and random-basis attacks), Eve’s information is quantified as the fraction of sifted key bits correctly inferred by Eve.

C. Detection Probability and Abort Thresholds

In practical QKD systems, Alice and Bob abort key generation if the observed QBER exceeds a predefined threshold. Motivated by theoretical security analyses of BB84, we adopt an abort threshold of 11

D. Statistical Stability and Averaging

All simulation-based experiments are repeated across multiple independent runs to account for stochastic variability arising from random bit generation, basis choices, and noise processes. We report mean values and standard deviations for QBER and Eve information, enabling quantitative comparison of attack strategies and parameter regimes. We evaluate each attack using consistent metrics, including QBER, Eve’s information gain, and detection probability relative to a predefined abort threshold.

VI. RESULTS AND ANALYSIS

This section presents an extensive analysis of experimental results obtained across the three layers of our study: classical Monte Carlo simulation, quantum circuit-level simulation using Qiskit Aer, and execution on real IBM Quantum hardware. The objective is not only to report numerical values of evaluation metrics, but also to interpret these results in the context of quantum information theory, practical QKD deployment, and adversarial detectability.

In the absence of any eavesdropping activity, baseline BB84 performance serves as a reference for protocol correctness and intrinsic noise. Classical Monte Carlo simulations yield near-zero QBER, confirming that the protocol logic and sifting procedures are correctly implemented. Quantum circuit-level simulations using Qiskit Aer similarly exhibit negligible QBER under ideal conditions, validating that gate-level modeling preserves the expected BB84 behavior.

In contrast, experiments conducted on real IBM Quantum hardware consistently exhibit non-zero baseline QBER, typically ranging between 5% and 15%. This deviation arises from a combination of gate infidelities, readout errors, decoherence, and crosstalk. Importantly, this baseline error establishes a noise floor that fundamentally limits the sensitivity of QBER-based eavesdropping detection. Any adversarial disturbance must be interpreted relative to this intrinsic hardware noise.

A. Baseline BB84 Performance

The baseline performance of the BB84 protocol establishes a reference against which all eavesdropping-induced disturbances must be evaluated. In the absence of any adversarial intervention, the protocol’s correctness depends solely on idealized quantum behavior and physical noise introduced by the transmission medium and hardware implementation.

Classical Monte Carlo simulations of BB84 yield near-zero quantum bit error rate (QBER), confirming the logical correctness of state preparation, basis selection, and sifting procedures. These results align with theoretical expectations, as no measurement-induced disturbance is present when Alice and Bob operate honestly in an ideal channel.

Quantum circuit-level simulations using Qiskit Aer similarly exhibit negligible QBER under ideal noise-free conditions. These results demonstrate that explicit quantum gate implementations preserve the fundamental behavior of BB84 and that circuit-level modeling faithfully reproduces the expected information-theoretic properties of the protocol.

In contrast, experiments executed on real IBM Quantum hardware consistently exhibit non-zero baseline QBER, typically in the range of 5% to 15%, even in the absence of an eavesdropper. This degradation arises from a combination of gate infidelities, readout errors, decoherence, and crosstalk inherent to current noisy intermediate-scale quantum (NISQ) devices. Importantly, this baseline error rate defines a noise floor that fundamentally limits the sensitivity of QBER-based eavesdropping detection.

As a consequence, any adversarial disturbance must be interpreted relative to this intrinsic noise. Attacks that introduce QBER comparable to the baseline noise level may evade detection, particularly under finite-size conditions. This observation motivates the subsequent comparative analysis of eavesdropping strategies and highlights the necessity of distinguishing protocol-level security guarantees from implementation-level limitations.

TABLE I
BASELINE BB84 PERFORMANCE ACROSS EXPERIMENTAL PLATFORMS.

Platform	QBER (Mean $\pm$ Std)	Eve Info	Detectable
Classical Simulation	$\approx 0.00 \pm 0.00$	None	No
Qiskit Aer (Ideal)	$\approx 0.00 \pm 0.01$	None	No
IBM Quantum Hardware	0.05–0.15	None	No

B. Intercept–Resend Attack

The intercept–resend attack represents the most aggressive and conceptually simple eavesdropping strategy. In both classical and circuit-level simulations, the observed QBER closely matches the theoretical value of 25%, arising from Eve’s random basis selection and the collapse of non-orthogonal quantum states. Eve’s information gain remains limited due to basis mismatch, despite the high disturbance introduced.

On real hardware, QBER frequently exceeds the theoretical 25% threshold due to additive hardware noise. Consequently, intercept–resend attacks are overwhelmingly detectable across all experimental layers, reaffirming their pedagogical value but limited practical stealth.

Random-basis attacks reveal a critical trade-off between stealth and information gain. Parameter sweeps demonstrate an approximately linear increase in QBER with attack probability p , while Eve’s information gain increases sublinearly. At low values of p , Eve can extract non-trivial information while keeping QBER below abort thresholds, particularly in noisy channels where hardware-induced errors mask adversarial disturbance.

These results highlight that partial eavesdropping strategies pose a more realistic threat than full intercept–resend attacks and underscore the limitations of QBER-only security analysis.

TABLE II
INTERCEPT-RESEND ATTACK METRICS (MEAN $\pm$ STANDARD DEVIATION).

Platform	QBER	Eve Info	Detection Prob.
Classical Simulation	0.25 ± 0.01	0.58 ± 0.02	≈ 1.0
Qiskit Aer	0.25 ± 0.03	0.56 ± 0.04	≈ 1.0
IBM Quantum Hardware	> 0.25	≈ 0.55	≈ 1.0

TABLE III
RANDOM-BASIS ATTACK METRICS ACROSS ATTACK PROBABILITIES.

Attack Probability p	QBER	Eve Information	Detectable
$p \approx 0.1$	Low (< 0.10)	Low-Medium	Rare
$p \approx 0.5$	Medium (~ 0.12)	Medium	Sometimes
$p \approx 1.0$	High (~ 0.25)	Medium	Yes

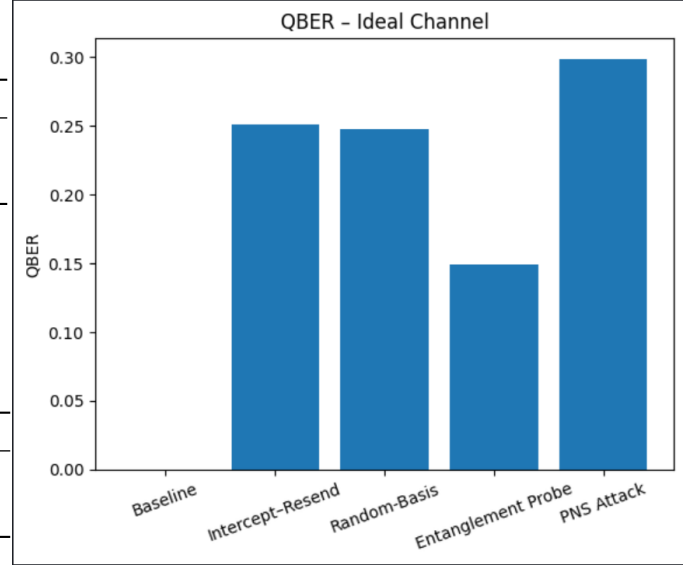


Fig. 1. Comparative QBER across all eavesdropping strategies using Monte Carlo Methods in Ideal Channels

C. Entanglement-Based Probe Attacks

Entanglement-based probe attacks exhibit a smooth and continuous trade-off between information gain and disturbance. Unlike intercept-resend, disturbance increases gradually with probe strength, enabling Eve to remain below detection thresholds while still acquiring partial information. Circuit-level simulations confirm this behavior, while hardware experiments show that intrinsic noise further obscures probe-induced disturbance.

These results demonstrate that entanglement-based attacks represent a more subtle and theoretically significant threat model, emphasizing the need for security analysis based on mutual information rather than simple error rates.

TABLE IV
ENTANGLEMENT-BASED PROBE ATTACK METRICS AS A FUNCTION OF PROBE STRENGTH.

Probe Strength	QBER	Eve Correlation	Detectable
Weak	Low (< 0.05)	Low	Rare
Moderate	Medium ($0.05-0.15$)	Medium	Sometimes
Strong	High (> 0.20)	High	Yes

D. Photon Number Splitting (PNS) Attacks

PNS attacks introduce negligible additional QBER while allowing Eve to obtain substantial information from multi-photon pulses. As a result, QBER-based detection completely fails against such attacks. These findings reinforce the distinction between protocol-level and implementation-level security and motivate the necessity of decoy-state techniques in practical QKD systems.

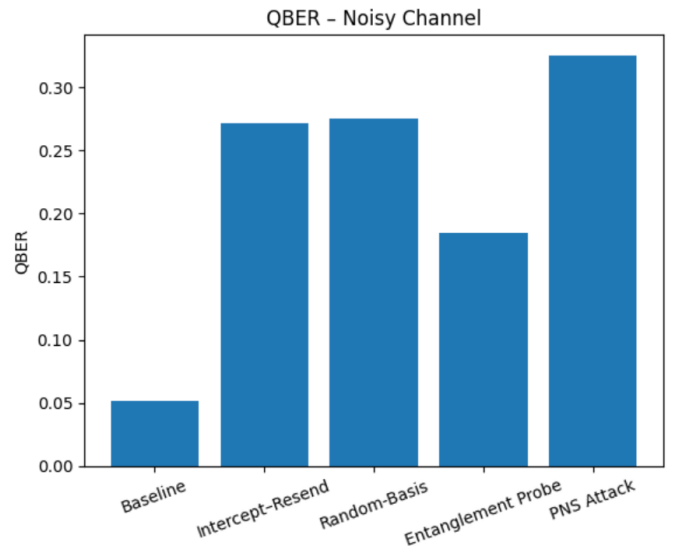


Fig. 2. Comparative QBER across all eavesdropping strategies using Monte Carlo Methods in Noisy Channels

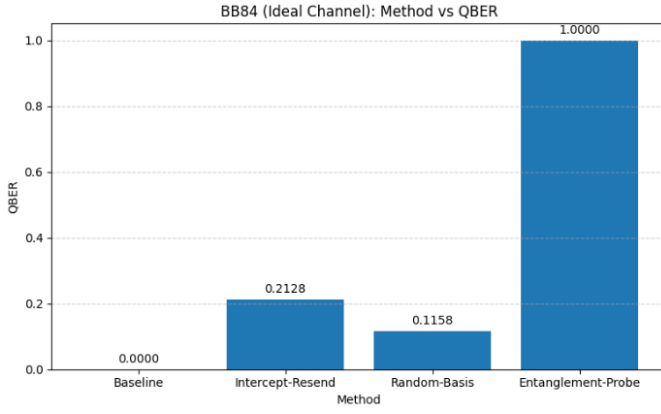


Fig. 3. Comparative QBER across all eavesdropping strategies using Qiskit Simulators in Ideal Channels

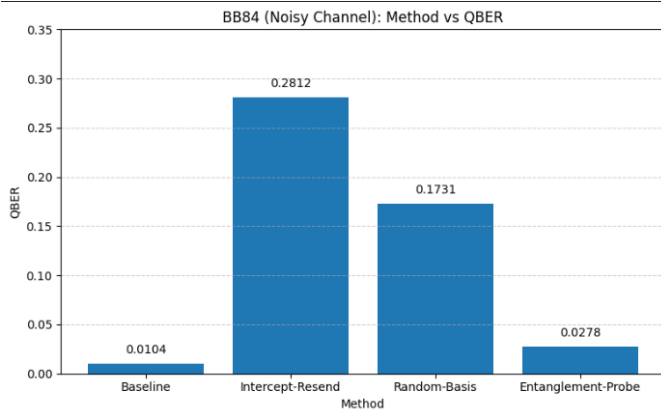


Fig. 4. Comparative QBER across all eavesdropping strategies using Qiskit Simulators in Noisy Channels

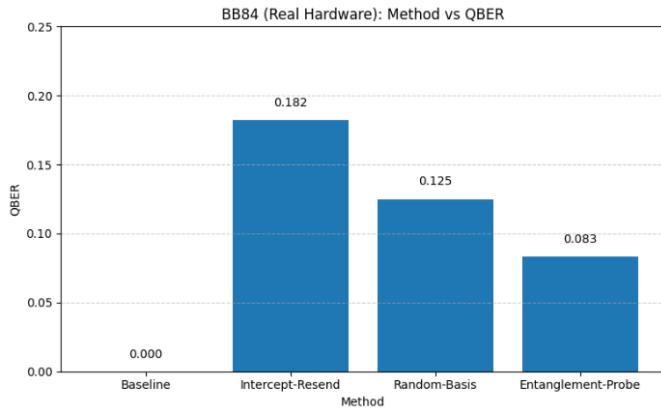


Fig. 5. Comparative QBER across all eavesdropping strategies using IBM Quantum Hardware

TABLE V
LIKELIHOOD OF DETECTION USING A QBER ABORT THRESHOLD OF 25%.

Attack Strategy	Detection Likelihood
Intercept-Resend	Very High
Random-Basis ($p > 0.5$)	Moderate
Random-Basis ($p < 0.2$)	Low
Entanglement Probe (Weak)	Very Low
Photon Number Splitting	None

VII. RELATED WORK

The security of BB84 has been extensively studied since its introduction. Bennett and Brassard first proposed the protocol and demonstrated its resistance to simple eavesdropping attacks. Fuchs and Peres analyzed optimal entanglement-based attacks, establishing fundamental information-disturbance trade-offs. L’utkenhaus and later Brassard *et al.* highlighted vulnerabilities arising from multi-photon emissions, leading to the development of decoy-state protocols. More recent work has focused on practical security analyses incorporating device imperfections and side channels. Our work complements these studies by providing a unified experimental comparison across simulation and real hardware.

VIII. ATTACKS BEYOND THE SCOPE OF THIS STUDY

While this work presents an extensive experimental analysis of several representative eavesdropping strategies against the BB84 protocol, there exist additional classes of attacks that are well-studied in the quantum cryptography literature but fall outside the scope of the present study. These attacks are excluded either due to their fundamentally different threat models, their reliance on implementation-specific vulnerabilities, or the significant analytical and computational complexity required for faithful modeling.

A. Optimal Individual Attacks

Optimal individual attacks, such as those analyzed by Fuchs and Peres, involve eavesdropping strategies that maximize Eve’s accessible information for a given level of disturbance by employing optimal positive-operator-valued measurements (POVMs). These attacks define tight information-disturbance trade-offs and serve as benchmarks in formal security proofs of BB84. Implementing such attacks requires explicit computation of mutual information or Holevo bounds, which is beyond the empirical metrics used in this study. Consequently, these attacks are discussed conceptually but not implemented experimentally.

B. Collective and Coherent Attacks

In collective attacks, Eve interacts individually with each transmitted qubit but defers measurement until after classical basis reconciliation, performing joint measurements across many stored quantum systems. Coherent attacks generalize this further by allowing Eve to interact with multiple qubits simultaneously. While these attacks are central to asymptotic security proofs, they are computationally intractable to simulate faithfully due to exponential Hilbert space growth and are

infeasible to realize in near-term quantum hardware. As such, they are excluded from the present experimental framework.

C. Side-Channel and Detector Attacks

Practical quantum key distribution systems are vulnerable to side-channel attacks that exploit imperfections in detectors, sources, and classical control systems. Examples include detector blinding attacks, timing attacks, and Trojan-horse attacks that inject auxiliary signals into legitimate devices. These attacks do not target the quantum channel itself and therefore violate the threat model assumed by BB84. Since the focus of this study is on quantum-channel eavesdropping strategies, side-channel attacks are not considered.

D. Man-in-the-Middle Attacks

The BB84 protocol assumes the existence of an authenticated classical communication channel between Alice and Bob. Man-in-the-middle attacks that exploit failures in classical authentication mechanisms undermine this assumption and represent classical cryptographic failures rather than quantum vulnerabilities. Such attacks are therefore explicitly excluded from this study.

By delineating these attack classes as out of scope, this work maintains a clear and consistent threat model while focusing on experimentally realizable and conceptually illustrative eavesdropping strategies. Future work may integrate formal security analysis or hardware-level countermeasures to address these broader classes of attacks.

IX. LIMITATIONS AND FUTURE WORK

This study is subject to several limitations. Finite-size effects and limited hardware shots restrict statistical confidence on real devices. Entanglement-based attacks are evaluated using simplified probe models, and full mutual information bounds are not computed. Future work includes implementing decoy-state BB84, performing Holevo-bound-based security analysis, and extending experiments to measurement-device-independent QKD and networked settings.

X. CONCLUSION

We presented a comprehensive experimental analysis of eavesdropping strategies against BB84 QKD across classical simulation, quantum circuit modeling, and real quantum hardware. Our results demonstrate that security depends critically on both attack models and implementation assumptions. By bridging theory and practice, this work provides valuable insight into the real-world security of quantum key distribution systems.

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TABLE VI
COMPARISON OF EAVESDROPPING STRATEGIES AGAINST BB84.

Attack Strategy	QBER Impact	Eve Info	Detectability	Security Scope
Baseline (No Eve)	None	None	No	Protocol
Intercept–Resend	High	Medium	Yes	Protocol
Random-Basis	Variable	Medium	Sometimes	Protocol
Entanglement Probe	Low–Medium	Partial	Rare	Protocol
Photon Number Splitting	None	High	No	Implementation